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ORIGINAL INVESTIGATION

Effect of Personalized Sodium Replacement on Fluid and Sodium Balance and Thermophysiological Strain During and After Ultraendurance Running in the Heat

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Purpose: To investigate the effect of personalized sweat sodium replacement on drinking behavior, sodium and water balance, and thermophysiological responses during and after ultraendurance running in hot conditions. ***Methods:*** Nine participants (7 male, 2 female) completed two 5-hour treadmill runs (60% maximum oxygen uptake, 30°C ambient temperature), in a double-blind randomized crossover design, consuming sodium chloride (SODIUM) capsules to replace 100% of previously assessed losses or placebo (PLACEBO). Fluid was consumed ad libitum. ***Results:*** No effect of SODIUM was observed for ad libitum fluid intake or net fluid balance ($P > .05$). Plasma sodium concentration increased in both trials, but to a greater extent in SODIUM at 2.5 hours (mean [SD]: 4 [4] mmol·L^{-1} vs 1 [5] mmol·L^{-1}; $P < .05$) and postexercise (4 [3] mmol·L^{-1} vs 1 [5] mmol·L^{-1}; $P < .05$). Plasma volume change was not different between trials ($P > .05$) but was strongly correlated with sodium balance in SODIUM ($r = .880$, $P < .01$). No effect of sodium replacement was observed for heart rate, rectal temperature, thermal comfort, perceived exertion, or physiological strain index. During the 24 hours postexercise, ad libitum fluid intake was greater following SODIUM (2541 [711] mL vs 1998 [727] mL; $P = .04$), as was urinary sodium excretion (NaCl: 66 [35] mmol, Pl: 21 [12] mmol; $P < .01$). ***Conclusions:*** Personalized sweat sodium replacement during ultraendurance running in hot conditions, with ad libitum fluid intake, exacerbated the rise in plasma sodium concentration compared to no sodium replacement but did not substantially influence overall body-water balance or thermophysiological strain. A large sodium deficit incurred during exercise leads to substantial renal sodium conservation postexercise.

Keywords: hydration, electrolytes, sweat sodium, exertional heat stress, personalized nutrition

Prolonged sweating during ultraendurance exercise could theoretically cause sodium losses that exceed 25% of total body sodium if unreplaced.[1] Although sodium losses are recognized in sport nutrition guidelines, quantifiable replacement strategies are lacking.[2,3] This is probably because, despite being commonplace, no studies have investigated personalized sodium replacement based on prior assessment. Given the substantial interindividual variation in sodium losses,[4] understanding the efficacy of personalized replacement is of great interest to athletes and practitioners and may provide insights previous studies have failed to observe due to arbitrary sodium replacement methods. Case reports suggest that high sodium intake during ultraendurance exercise may inappropriately stimulate thirst, causing excessive ad libitum fluid consumption that paradoxically increases risk of exercise-associated hyponatremia.[3] Some authors have also suggested osmotically inactive sodium stores could be released into circulation in response to reduced plasma osmolality (P_{Osm}), supporting an argument that quantifiable sodium replacement is unnecessary even in ultraendurance exercise.[5] These viewpoints, however, lack evidence beyond isolated case examples and published hypotheses. The only laboratory-controlled studies of sodium replacement during ultraendurance exercise (duration ≥ 4 h) provided fixed, complete water replacement,[6,7] an approach that is neither commonly practiced nor recommended.[2,3]

Postexercise, data suggest urinary sodium excretion (U_{Na}) is substantially reduced in response to an exercise-induced sodium deficit.[8,9] The effect of a larger deficit on sodium balance and hydration status following ultraendurance exercise has not been previously studied but is of particular relevance for multiday events. With these considerations in mind, the aim of this study was to determine the effect of personalized sodium replacement during ultraendurance running in the heat, on: (1) thirst and ad libitum fluid intake, (2) water and sodium balance, (3) thermophysiological strain during exercise, and (4) presenting hydration status 24 hours postexercise. We hypothesized that compared with no intake, sodium replacement would: (1) increase ad libitum fluid intake and net water balance during and postexercise (2) increase postexercise plasma sodium ($[Na^+]_{plasma}$), and P_{Osm}, (3) reduce heart rate (HR) and rectal temperature (T_{re}), but not rating of perceived exertion (RPE) or thermal comfort rating (TCR) during exercise, (4) reduce postexercise U_{Na} following exercise with no sodium replacement, but (5) this would not influence hydration status 24 hours postexercise.

Methods

Participants and Ethical Approval

Nine "recreationally active" or "trained"[10] ultraendurance runners (7 male, 2 female) volunteered for this study (mean [SD]: age 38 [9] y, height 172 [4] cm, body mass [BM] 66.4 [8.2] kg, $\dot{V}O_2max$ 55.7 [3.9] mL·kg^{-1}·min^{-1}), which conformed to the standards of the Declaration of Helsinki and was approved by the Monash University Research Ethics Committee (Project # 18899).

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Exclusion criteria included medical conditions impacting renal, sweat, or thermoregulatory function, injuries, or conditions preventing exercise completion. Participants provided written consent prior to participation. Full participant characteristics are provided in Table S1 in the Supplementary Material (available online).

Preliminary Measures and Familiarization

Participants attended the laboratory 7 to 14 days before the first experimental trial. Height, BM, fat, and fat-free mass were measured (Seca 515 MBCA, Seca Group). Maximum oxygen uptake ($\dot{V}O_2$max) was estimated using breath-by-breath indirect calorimetry (Vyntus CPX, Vyaire) during a continuous, incremental treadmill test as previously described,[11] with $\dot{V}O_2$max taken as the greatest 30-second average $\dot{V}O_2$ once respiratory exchange ratio had exceeded 1.0, $\dot{V}O_2$ had plateaued, and the participant had reached volitional exhaustion. Running speed eliciting 60% $\dot{V}O_2$max (8.7 [0.6] km·h^{-1}) was confirmed with the same system, before completing a 1-hour familiarization run at this speed, in a climate chamber at ambient temperature (T_{amb}) 30.3 (0.8)°C, relative humidity (RH) 32.2% (3.5%), and fan airspeed ~10.5 km·h^{-1}. Familiarization was used to collect sweat samples for personalizing sodium replacement in experimental trials. Sterile absorbent patches were applied (Tegaderm + Pad, 6 × 10 cm, 3M Health) to the ventral forearm and midthigh. Skin preparation, patch application, removal, processing, and analysis were completed as previously described.[12] Briefly, the skin sites were cleaned with an alcohol wipe, rinsed with deionized water, and dried with a clean laboratory wipe (Kimberly-Clark) prior to patch application. Patches were removed when approximately 25% of the absorbent area was saturated with sweat (approximately 30–45 min for forearm patches, and 30–60 min for midthigh patches). Whole-body $[Na^+]_{sweat}$ was estimated using published regression equations,[13] and data combined with whole-body sweat rate, estimated from BM balance, to calculate whole-body sodium loss. Participants consumed water ad libitum and completed measures of RPE on a 6 to 20 scale,[14] TCR on a 12-point Likert-type scale,[15] and perceived thirst and salt craving on a 0 to 10 scale.[16,17]

Experimental Procedure

Participants completed 2 experimental trials in randomized order, replacing 100% (SODIUM) or 0% (PLACEBO) of expected sweat sodium losses based on the familiarization. 100% sodium replacement was chosen because many athletes and sweat testing companies advocate this approach, and to ensure any effect of sodium on thirst and ad libitum fluid intake could be observed. Each trial period lasted 5 days (Figure 1), with participants consuming pre-prepared food (energy: 185 kJ·kg^{-1}·d^{-1}, protein: 1.5 g·kg^{-1}·d^{-1}, carbohydrate [CHO]: 6 g·kg^{-1}·d^{-1}, sodium: 50 mg·kg^{-1}·d^{-1}), and collecting all urine produced. The sodium intake level was chosen to closely resemble that observed previously in endurance athletes eating their usual diet.[12] Participants refrained from sweat-inducing activities throughout the trial period, except for the run on day 4. Trials were timed to consecutive follicular phases of the menstrual cycle for females with a regularly occurring cycle, confirmed from plasma estradiol (DKO003/RUO; DiaMetra) concentration (all <50 pg·mL^{-1}, between-trial variation 24 pg·mL^{-1}), or otherwise separated by 1 to 2 weeks.

On day 4, participants consumed a standardized breakfast (CHO: 1.5 g·kg^{-1}, protein: 0.25 g·kg^{-1}, fluid: 250 mL, Na: 420 mg) 2 hours prior to running. Thirty minutes prior to running, after bladder voiding, total body water (TBW) and extracellular water (ECW) were assessed (Seca 515 MBCA, Seca). After sitting for 5 minutes, blood was taken by venipuncture from the antecubital vein into a vacutainer (6 mL, 1.5 IU·mL^{-1} lithium heparin). Participants then measured preexercise nude BM before inserting a thermocouple 12 to 15 cm beyond the anal sphincter (Alpha-Technics 4600) for measuring T_{re}. The running protocol replicated the familiarization, but for 5 hours duration. During exercise, T_{re} was recorded every 15 minutes, and RPE, HR, TCR, thirst, and salt craving every 30 minutes. Physiological strain index[18] was calculated during analysis, representative of the combined thermoregulatory and cardiovascular strain. After 30-minute running, and in the final hour, sweat patches were applied as previously described to quantify sweat sodium losses in the first and fifth hour of exercise. After 2.5-hour running, participants briefly left the climate chamber for blood sampling as described previously.

Immediately postexercise, nude BM, after toweling dry, was measured, followed by postexercise blood sampling. Participants continued dietary standardization and urine collection and were provided three 1.5 L water bottles, from which all additional fluids were consumed until returning to the laboratory. Bottles were weighed prior to leaving and on return to quantify ad libitum fluid intake postexercise. Participants returned to the laboratory on day 5, having consumed the same breakfast, at the same time as day 4. 24 hours after preexercise measurements, final nude BM, TBW, ECW, and blood samples were obtained.

CHO, Fluid, and Sodium Provision During Exercise

Throughout exercise participants consumed an unflavored solution providing 1 g·kg^{-1}·h^{-1} CHO (20% w/v).[19] This fixed-volume solution replaced 34% (4%) of participants' water losses. A second bottle of plain water was provided for additional ad libitum intake. This bottle was weighed at the beginning and end of each hour and replaced. Both bottles were within reach of participants at all times. SODIUM or PLACEBO was provided in opaque, methylcellulose capsules. SODIUM capsules were filled with NaCl to match the rate of sweat sodium loss calculated from the familiarization, or the equivalent mass of maltodextrin (PLACEBO). The number of capsules given was divisible by 5, allowing consistent intake across each hour of running. Participants and researchers were blinded to capsule contents until after data analysis. All participants consumed food, capsules, and fluids as intended.

Calculation of Sweat Loss, TBW Change, and Sodium Loss During Running

Total sweat loss and Δ TBW during exercise were calculated from the change in BM, corrected as described by Hoffman et al,[20] with the exception of glycogen water release, which has since been shown not to constitute an additional source of water input.[21] Whole-body sodium loss was estimated from the corrected total sweat loss and estimated whole-body $[Na^+]_{sweat}$ obtained during the first and final hour of running, with the final hour value used as an approximation of hours 2 to 4, given that $[Na^+]_{sweat}$ increases during the initial hour of exercise.[22]

Sweat, Urine, and Blood Analysis

Sweat and urine sodium were determined using ion-selective electrodes (L'AQUATwin Na-11, Horiba) as previously described.[12] Hemoglobin (Hb201+, Hemocue AB) and hematocrit

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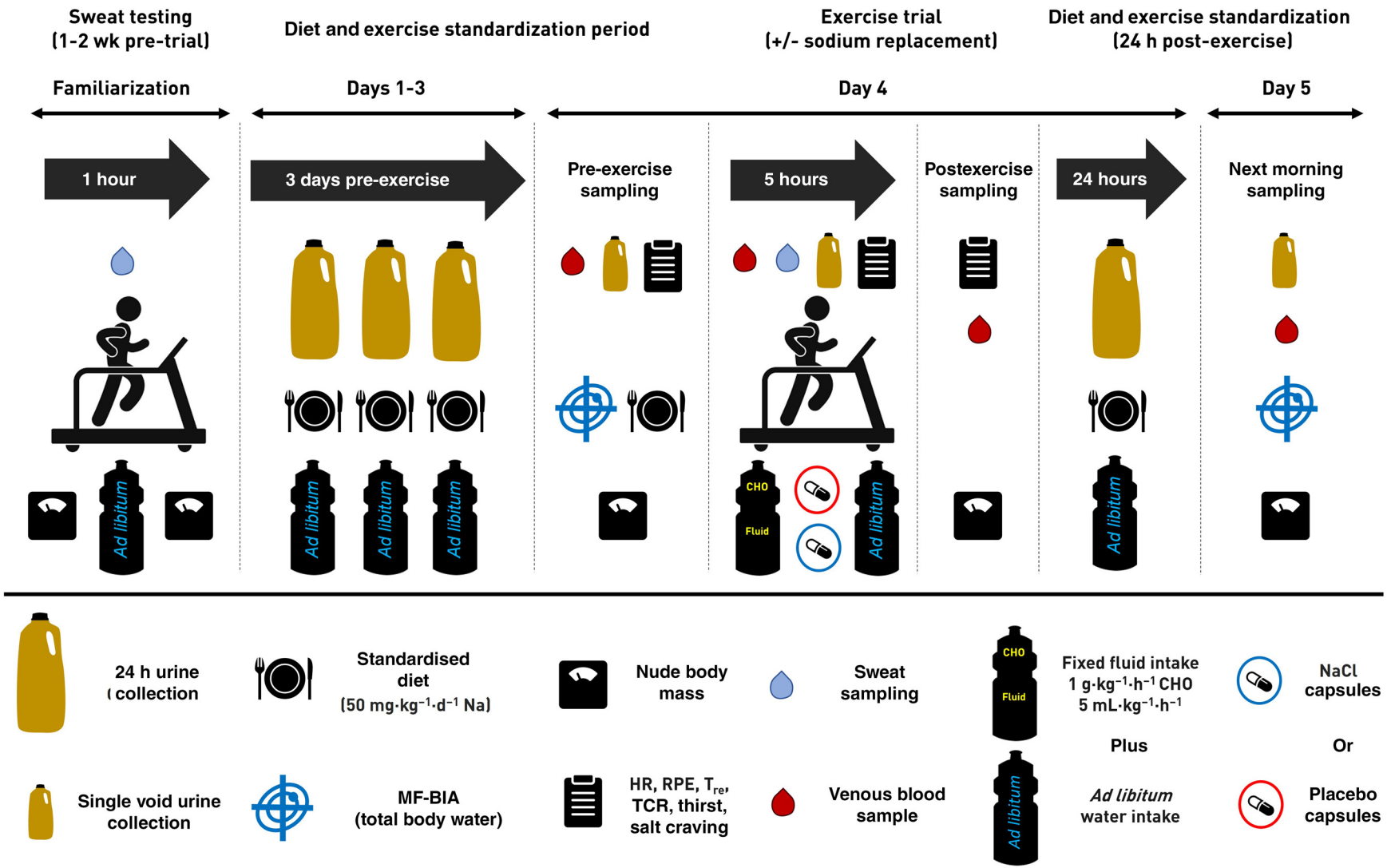

Figure 1 — Schematic description of the experimental design. CHO indicates carbohydrate; HR, heart rate; MF-BIA, multifrequency bioelectric impedance analysis; Na, sodium; RPE, rating of perceived exertion; T_{re}, rectal temperature; TCR, thermal comfort rating.

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(centrifuged capillary tubes, Propper) were measured from whole blood and used to determine plasma volume (ΔP_v).[23] Blood was then centrifuged (4000 RPM, 10 min, 4 °C) and the resultant plasma used to measure P_{Osm} using freeze-point osmometry (Osmomat 030; Gonotec). The remainder was frozen at −80 °C until analysis. $[Na^+]_{plasma}$ was assessed (Cobas c501, Roche Diagnostics) by external pathology services (Cabrini Pathology). Plasma cortisol was assessed for the stress response to exercise, aldosterone for sodium homeostatic regulation, and Vascular Endothelial Growth Factor C (VEGF-C) as a marker of osmotically inactive Na^+ release.[24] Cortisol (DKO001, DiaMetra), aldosterone (DKO053, DiaMetra), and VEGF-C (ELH-VEGFC-1, RayBiotech) were measured in duplicate using enzyme-linked immunosorbent assay (ELISA), with participants' samples on the same plate, and standards and controls on all plates. Coefficients of variation for cortisol, aldosterone, and VEGF-C were 3.7%, 5.3%, and 5.0%, respectively.

Statistical Analysis

Using previously published data[12,25] and a standard alpha (.05) and beta value (0.20), a sample size of 8 was calculated to provide adequate statistical power to detect differences between trials for $[Na^+]_{plasma}$ of 5 mmol·L^{-1}, ΔP_v of 10%, ad libitum fluid intake of 300 mL·h^{-1}, and postexercise U_{Na} of 50% (G*Power 3.1). Data are presented as mean (SD) unless indicated, and error bars on figures as SEM for clarity. Data distribution was checked for normality using the Shapiro–Wilk test before analysis. Multiple timepoint data were analyzed using 2-way, repeated-measures analysis of variance and presented with effect size (partial eta squared, η_p^2). Homogeneity and sphericity were checked, adjustments to degrees of freedom made using Greenhouse–Geisser correction where necessary, and pairwise comparisons made using Bonferroni correction. Single timepoint data were analyzed using paired-sample *t* tests or nonparametric equivalent (Wilcoxon signed-rank). Pearson correlation was performed between ΔP_v and both net water and sodium balance. All analyses were performed using SPSS (version 28.0), with statistical significance accepted at $P < .05$.

Results

Trial Completion and Blinding

All participants completed all trials and confirmed consumption of the pre-prepared food and fluids. All sodium capsules and provided fluids were consumed as intended during exercise. At completion, 4 participants correctly guessed trial order, 3 incorrectly guessed trial order, and 2 were unable to distinguish and declined to guess. No effect of trial order was observed for any variable (data not shown).

Sodium Balance and Salt Craving

Whole-body $[Na^+]_{sweat}$ and sodium loss during familiarization were 50 (14) mmol·L^{-1} and 1217 (538) mg·h^{-1}, respectively. There were no differences in U_{Na} on days 1 to 3 between SODIUM (32 [13] mg·kg^{-1}·d^{-1}) and PLACEBO (31 [9] mg·kg^{-1}·d^{-1}, $P = .67$). During SODIUM, capsules replaced 117% (49%) (range 78%–231%) of sweat sodium losses, with most participants (n = 5) replacing within 20% of actual losses. As intended, a large net sodium deficit occurred during PLACEBO (−250 [102] mmol) but not SODIUM (+14 [114] mmol, $P < .001$). Salt craving increased throughout exercise ($P = .01$, $\eta_p^2 = .24$) but was not different between trials. Postexercise U_{Na} was significantly reduced following PLACEBO (21 [12] mmol) compared to SODIUM (66 [35] mmol, $P < .01$); however, net sodium balance from preexercise to 24 hours postexercise remained negative in PLACEBO (−133 [95] mmol) and positive in SODIUM (87 [108] mmol; $P < .01$).

Water Balance and Hydration Status

Preexercise hydration markers are presented in Table 1, with no differences between trials other than 0.3 L greater ECW prior to SODIUM ($P = .04$). During exercise, thirst rating increased over time ($P < .001$, $\eta_p^2 = .6$) but was not different between trials (Figure 2E). Likewise, no significant difference in ad libitum water intake was observed between trials ($P = .07$). Water loss and net water balance during exercise also did not significantly differ between trials (Table 2). There was no difference in urine volume postexercise, but ad libitum water intake was greater following SODIUM, resulting in greater postexercise water balance (1532 [507] mL vs 774 [588] mL, $P < .001$).

Data for $[Na^+]_{plasma}$, P_{Osm}, and ΔP_v are presented in Figure 3. A time × trial interaction was observed for $[Na^+]_{plasma}$ ($P = .04$, $\eta_p^2 = .29$), with $[Na^+]_{plasma}$ significantly greater in SODIUM than PLACEBO at 2.5 and 5 hours. P_{Osm} increased in both trials (main effect of time; $P < .01$, $\eta_p^2 = .46$), with significantly greater P_{Osm} compared with preexercise at 5 hours only ($P < .05$). P_v changes between trials were not statistically significantly different ($P = .07$). A strong correlation was observed between net sodium balance and ΔP_v in SODIUM ($r = .88$, $P < .01$) but was not significant for PLACEBO ($r = .63$, $P = .07$) (see Figure S1 in the Supplementary Material [available online]). No correlation was observed between net fluid balance and ΔP_v in either trial (SODIUM: $r = .26$, $P = .50$; PLACEBO: $r = .49$, $P = .19$) (see Figure S2 in the Supplementary Material [available online]).

Table 1 Preexercise Sodium Excretion and Hydration Status

	Sodium	Placebo	*P*
Urinary Na^+ excretion, mg·kg^{-1}·d^{-1}	32 (13)	31 (9)	.67
First-void urine-specific gravity	1.019 (0.007)	1.019 (0.006)	.78
Total body water, L	38.4 (4.2)	38.1 (3.9)	.05
Extracellular water, L	16.2 (2.0)	15.9 (1.8)	.04
Intracellular water, L	23.3 (3.0)	23.3 (2.8)	.65
Plasma sodium, mmol·L^{-1}	141 (3)	143 (5)	.21
Plasma osmolality, mOsm·kg^{-1}	297 (5)	299 (4)	.32

Note: Values are presented as mean (SD).

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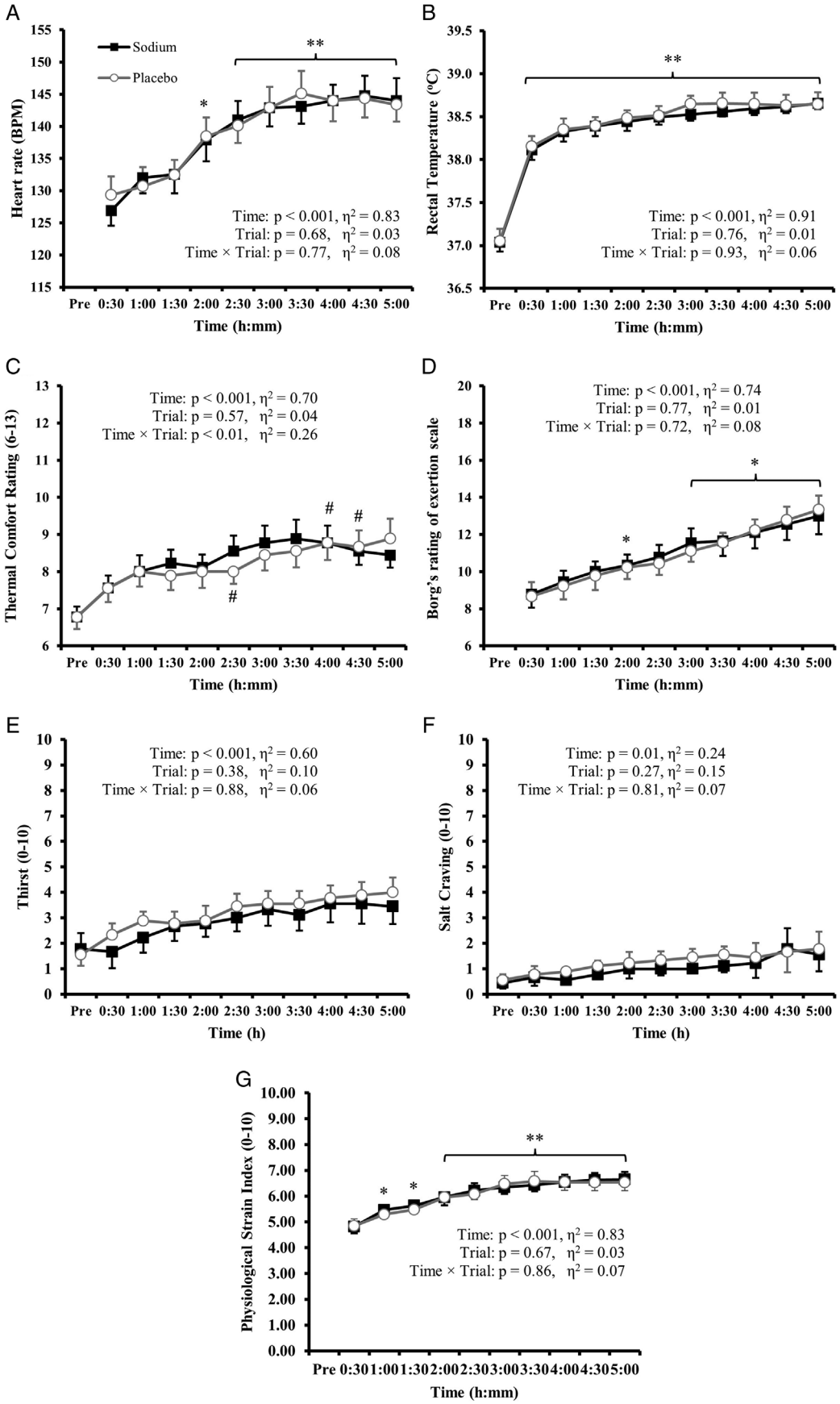

Figure 2 — Heart rate (panel A), rectal temperature (B), thermal comfort rating (C), rating of perceived exertion (D), thirst perception (E), salt craving (F), and physiological strain index (G) during 5-hour steady-state running exercise at 60% maximal oxygen uptake in hot (30 °C) ambient conditions. SODIUM (■), PLACEBO (○). Mean ± SEM (n = 9): $^{*}P < .05$ versus first time point. $^{**}P < .01$ versus first time point. $\#P < .05$ versus first time point (PLACEBO only). PLACEBO indicates replacing 0% of expected sweat sodium loss; SODIUM, replacing 100% of expected sweat sodium loss.

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Table 2 Water and Sodium Balance During and After 5-Hour Steady-State Running Exercise at 60% $\dot{V}O_2$max in Hot (30 °C) Ambient Conditions

	During exercise			Postexercise			Total		
	Sodium	Placebo	*P*	Sodium	Placebo	*P*	Sodium	Placebo	*P*
Water loss, mL									
Sweat	4182 (689)	4207 (664)	.85	—	—	—	—	—	—
Respiratory	448 (66)	482 (77)	.24	—	—	—	—	—	—
Urine	299 (201)	240 (115)	.29	1009 (440)	1224 (855)	.28	1309 (465)	1464 (868)	.47
Total	4930 (748)	4929 (711)	.99	1009 (440)	1224 (855)	.28	5939 (1023)	6153 (1190)	.28
Fluid intake, mL									
Fixed	1662 (201)	1662 (201)	—	250 (0)	250 (0)	—	1912 (201)	1912 (201)	—
Ad libitum	1553 (506)	1216 (540)	.07	2291 (711)	1748 (727)	.04	3844 (962)	2964 (967)	.05
Total	3215 (552)	2878 (500)	.07	2541 (711)	1998 (727)	.04	5756 (1091)	4877 (989)	.05
Metabolic water production, mL	448 (61)	464 (50)	.09	—	—	—	—	—	—
Net water balance, mL	−1266 (766)	−1586 (900)	.10	+1532 (507)	+774 (588)	<.01	+265 (645)	−812 (622)	<.01
Water loss replaced, %	75 (12)	69 (14)	.10	285 (120)	286 (364)	.11	105 (11)	88 (11)	<.01
TBW loss, % of initial BM	1.9 (1.1)	2.4 (1.3)	.16	—	—	—	—	—	—
Sodium loss, mmol									
Sweat	224 (65)	235 (100)	.57	—	—	—	—	—	—
Urine	19 (20)	12 (7)	.26	66 (35)	21 (12)	.005	84 (50)	33 (12)	.01
Total	243 (69)	246 (102)	.85	66 (35)	21 (12)	.005	309 (77)	267 (95)	.06
Sodium intake, mmol	265 (117)	0 (0)	—	138 (20)	138 (20)	—	403 (124)	138 (20)	<.001
Net sodium balance, mmol	+14 (114)	−250 (102)	<.001	+73 (40)	+117 (25)	<.01	+87 (108)	−133 (95)	<.01
Sweat sodium loss replaced, %	117 (48)	0 (0)	—	—	—	—	—	—	—

Abbreviations: BM, body mass; TBW, total body water; $\dot{V}O_2$max, maximal oxygen uptake. Note: n = 9. The values are presented as mean (SD). Net water balance is calculated as total water gains (water intake [via foods and fluids] + metabolic water production) – total water loss (sweat, respiratory, and urine). Water loss replaced (%) is calculated as total water gain (fluid intake + metabolic water production) as a proportion of total water loss (sweat, respiratory, and urine).

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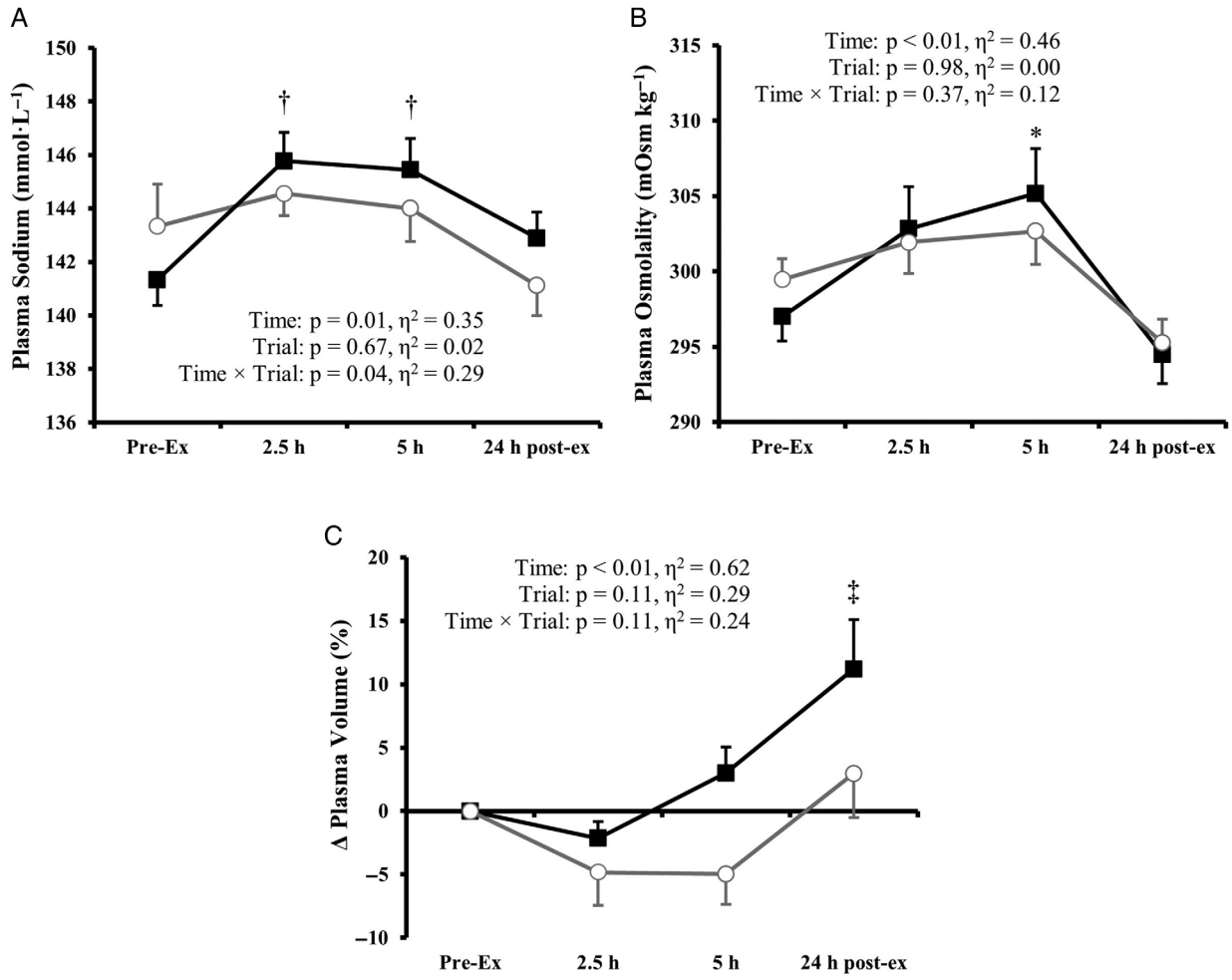

Figure 3 — (A) Plasma sodium concentration, (B) plasma osmolality, and (C) plasma volume before, during, and after 5-hour steady-state running exercise at 60% $\dot{V}O_2$max in hot (30°C) ambient conditions. SODIUM (■), PLACEBO (○). Mean ± SEM (n = 9): $^*P < .05$ versus first time point. †$P < .05$ versus first time point (SODIUM only). ‡$P < .05$ versus 2.5 hours (SODIUM only). SODIUM indicates replacing 100% of expected sweat sodium loss; PLACEBO, replacing 0% of expected sweat sodium loss; $\dot{V}O_2$max, maximal oxygen uptake.

Next morning, there was a reduction in ECW in PLACEBO and minimal change in SODIUM (SODIUM + 0.1 [0.5] L, PLACEBO −0.7 [0.6] L; $P = .02$), and no changes in intracellular water (ICW) (SODIUM + 0.1 [0.4] L, PLACEBO + 0.2 [0.2] L; $P = .27$).

Thermophysiological Strain

Results for HR, RPE, T_{re}, TCR, and physiological strain index during exercise are presented in Figure 2. Main effects of time were observed for each variable ($P \leq .01$), but no effect of trial was observed for any, and a time × trial interaction was observed only for TCR ($P < .01$, $\eta_p^2 = .26$), with pairwise comparisons indicated in the figure.

Plasma Cortisol, Aldosterone, and VEGF-C Concentrations

Plasma cortisol, aldosterone, and VEGF-C are presented in Table 3. A main effect of time but not trial was observed for plasma cortisol, and no interactions. Plasma aldosterone increased significantly compared with preexercise at 2.5 hours ($P < .05$), 5 hours ($P < .01$), and 24 hours ($P < .01$), but no significant effect of trial or trial × time interaction was observed. No differences were observed for VEGF-C at any time point.

Discussion

This study provided complete, personalized sodium replacement to investigate its effect on thirst and ad libitum fluid intake during ultraendurance exercise, and implications for body water, sodium balance, and thermo-physiological responses. Contrary to our hypothesis, personalized sodium replacement did not influence ad libitum fluid intake or net body water balance during exercise. While no significant difference between SODIUM and PLACEBO for ΔP_v was observed, a positive correlation existed between net sodium balance and ΔP_v during exercise in SODIUM. $[Na^+]_{plasma}$ was greater at the conclusion of exercise in SODIUM compared to PLACEBO; however, this difference resulted from elevated $[Na^+]_{plasma}$ in SODIUM rather than reduced $[Na^+]_{plasma}$ in PLACEBO. Sodium replacement did not influence thermophysiological strain throughout the 5 hours of steady-state running. Postexercise, a substantial reduction in U_{Na} was observed in PLACEBO as hypothesized, although this failed to completely compensate for the exercise-induced sodium deficit, and

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Table 3 Plasma Cortisol, Aldosterone, and VEGF-C Concentration in Response to 5-Hour Steady-State Running Exercise at 60% $\dot{V}O_2$max in Hot (30 °C) Ambient Conditions

	Sodium				Placebo				Main effects		
	Pre-exercise	2.5 h	Post-exercise	24 h post-exercise	Pre-exercise	2.5 h	Post-exercise	24 h post-exercise	Effect of time P (η_p^2)	Effect of trial P (η_p^2)	Time × trial interaction P (η_p^2)
Cortisol, ng·mL^{-1}	206 (63)	230 (85)	407 (194)	190 (98)	214 (74)	213 (64)	442 (170)	194 (108)	<.01 (.62)	.71 (<.01)	.66 (.03)
Aldosterone, pg·mL^{-1})	84 (52)	219 (140)	274 (269)	199 (174)	66 (49)	345 (249)	532 (289)	261 (175)	<.001 (.72)	.15 0.24)	.07 (.30)
VEGF-C, pg·mL^{-1}	493 (663)	502 (702)	560 (820)	642 (994)	488 (659)	498 (679)	509 (673)	574 (778)	.10 (.30)	.19 (.20)	.44 (.08)

Abbreviations: VEGF-C, vascular endothelial growth factor C; $\dot{V}O_2$max, maximal oxygen uptake. Note: n = 9. The values are presented as mean (SD).

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sodium and body water balance the following morning remained negative in PLACEBO, but positive in SODIUM.

This study is only the third laboratory-based study to investigate sodium replacement during exercise ≥4-hour duration, and the first to investigate both personalized sodium replacement and ad libitum fluid intake. Sanders et al[6] (n = 6) studied 4-hour cycle ergometry at 55% $\dot{V}O_2$max (~20 °C T_{amb} and 71% RH), replacing ~100% fluid losses with either 5 mmol·L^{-1}, 50 mmol·L^{-1}, or 100 mmol·L^{-1} sodium solution. Sodium overreplacement (~265% in the 100 mmol·L^{-1} trial) expanded ECW at the expense of ICW. Moderate sodium replacement (~130%) in the 50 mmol·L^{-1} trial maintained stable ECW at the expense of ICW, but when sodium was underreplaced (~13% in the 5 mmol·L^{-1} trial), ICW was maintained at the expense of ECW. While we were unable to quantify ECW and ICW changes during exercise, the significant positive correlation between net sodium balance and ΔP_v in SODIUM supports this finding. It is important to note that blood samples in our study were taken after only 5 minutes of seated rest. This is likely to result in increased values for hemoglobin, hematocrit, and P_{Osm}.[26] The timing of blood sample collection in this study design is somewhat of a compromise, since a longer rest interval (eg, 15 min) at the midway point of exercise (ie, 2.5 h) would likely impact on measured plasma cortisol, as well as the T_{re} and TCR responses when resuming running due to a longer rest period outside the environmental chamber. In either case, the consistency of the rest interval used across timepoints would suggest that the change in P_v and P_{Osm} between timepoints would be of a similar direction and magnitude, albeit absolute P_v would be greater and P_{Osm} lower with a longer rest interval. Barr et al[7] (n = 8) studied 6-hour cycle ergometry at 55% $\dot{V}O_2$max (~30 °C T_{amb} and 50% RH) with replacement of ~100% fluid losses through plain water, the same volume of a 25 mmol·L^{-1} sodium solution, or no fluid intake. The sodium trial replaced ~96% of sweat sodium losses, and similar to our study and the 50 mmol·L^{-1} trial of Sanders et al,[6] sodium replacement had minimal impact on fluid balance and sodium losses during exercise. In all 3 studies, no differences in HR or T_{re} were observed, regardless of trial conditions or changes in ECW or P_v, except when no fluid replacement was permitted.[7] These findings suggest that TBW, rather than its distribution between compartments, is more consequential for thermophysiological variables, and that differences in ECW or P_v resulting from sodium replacement over 4 to 6 hours are insufficient to influence these variables to a significant extent.

Guidelines for hydration in ultraendurance exercise have cautioned against sodium replacement in excess of losses, based on concerns from case studies that increased osmotically driven thirst could cause excessive fluid intake, and counterintuitively exercise-associated hyponatremia.[3] The present study found no evidence that slightly excessive sodium replacement (117% [49%] replacement, or 1217 [539] mg·h^{-1}) increased perceived thirst or ad libitum fluid intake during exercise to a substantial degree, with a nonsignificant difference in fluid intake of just 68 mL·h^{-1} between trials. It is possible that prompting participants to rate thirst and salt craving every 30 minutes moderated drinking behavior in SODIUM, although this is not reflected in thirst ratings. It is also possible that 5 hours is insufficient to see accumulation of subtle fluid excesses that may become important after 15 hours exercise. Sodium was provided in capsules to ensure blinding, and it is conceivable that sensing sodium in the upper gastrointestinal tract or salty taste in the mouth may influence thirst perception and/or drinking behavior in a way that was undetectable with our methodology.

We also sought to observe evidence for release of osmotically inactive sodium stores during ultraendurance exercise. Such a mechanism has been suggested to minimize the reduction in $[Na^+]_{plasma}$ and P_{Osm} when exogenous sodium is not consumed during exercise, providing some protection against exercise-associated hyponatremia.[5] The change in net sodium balance and ECW from before exercise to 24 hours later would predict an increase in $[Na^+]_{plasma}$ of 4.4 mmol·L^{-1} in SODIUM and a reduction of 2.2 mmol·L^{-1} in PLACEBO, if one made assumptions that $[Na^+]_{plasma}$ and $[Na^+]_{ECW}$ are equivalent, and no exchange of osmotically inactive sodium took place. The measured change in $[Na^+]_{plasma}$ in PLACEBO closely matches the abovementioned estimate (reduction of 2 mmol·L^{-1}), but in the case of SODIUM, the measured increase was only half of what was predicted. VEGF-C has been proposed as a biomarker of release of sodium from the osmotically inactive stores back into circulation through the lymphatic system.[27] Contrary to the hypothesis mentioned above, theoretical and clinical studies suggest that sodium release occurs in response to reduced TBW or P_v rather than P_{Osm}.[28–30] To the best of our knowledge, this study is the first to assess plasma VEGF-C responses to substantial exercise-induced changes in water and sodium balance. We found no intraindividual differences in VEGF-C throughout exercise in either trial, but very large interindividual variation, as has been reported previously in clinical populations.[31] Whether or not this should be interpreted to mean that no exchange of osmotically inactive sodium stores took place cannot be easily determined, although the above estimates would suggest that there was likely no need for such action to occur. This could be considered a likely explanation, because ΔTBW and ΔP_v were small (<1.6 L and <5%, respectively) and $[Na^+]_{plasma}$ and P_{Osm} increased during exercise regardless of sodium intake. However, there are other possibilities that could explain why no change in plasma VEGF-C was observed at any timepoint in either trial. First, our estimates of sodium and water balance during exercise are based on indirect measures (ie, corrected BM change and estimated sodium loss from regional sweat collections, respectively), which although validated will not perfectly reflect these changes. And second, previous studies investigating VEGF-C in response to disturbances in fluid and/or sodium balance have focused on clinical conditions that disrupt the typical regulation of these processes. It is therefore possible that hydration-related changes in ECW, P_v, and P_{Osm} are mediated through other pathways for osmotically inactive sodium that do not involve VEGF-C. Field studies at ultraendurance events, where greater variance in postexercise P_{Osm} and P_v changes would be expected, may provide a more optimal setting to test the utility of VEGF-C as a suitable marker of osmotically inactive sodium exchange during and postultraendurance exercise.

The finding of substantially reduced postexercise U_{Na} in response to an exercise-induced sodium deficit, independent of urine volume, reinforces the case study findings of Lichton[8] and the recent findings of Funnell et al.[9] In our study, a sodium deficit of 250 mmol during 5-hour running in PLACEBO was partially offset postexercise by U_{Na} that was 45 mmol lower than SODIUM. While this conservation of sodium by the kidneys was unable to completely restore sodium balance within 24 hours in the absence of increased sodium intake, this finding does suggest that targeting 100% replacement of an exercise-induced sodium deficit postexercise is unnecessary to restore sodium balance by the next day. Furthermore, the negative sodium balance postexercise will likely induce adaptations that reduce sweat and U_{Na} in subsequent days, so a large sodium deficit is

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unlikely to accumulate during multistage ultraendurance events.[12,32] Increased sodium intake postexercise may drive increased thirst, ad libitum fluid intake and water retention, although mixed meals containing a variety of nutrients likely have the same or similar effect.[33]

Our findings also demonstrate that precise, personalized sodium replacement is actually difficult to achieve. We provided sodium capsules based on losses measured during familiarization, but in some participants found large discrepancies in sodium loss rates between familiarization and experimental trials. Review of these discrepancies at the individual level showed that differences in $[Na^+]_{sweat}$ between familiarization and SODIUM were a contributor, whereby participants with greater sodium losses in familiarization also had greater $[Na^+]_{sweat}$ and vice versa. A potential explanation for this is lack of dietary sodium control prior to the familiarization (unlike experimental trials), which can impact $[Na^+]_{sweat}$ if the discrepancy in sodium intake is large enough.[12] In addition to this, however, differences in sweat rate between familiarization and experimental trials also contributed in some participants. In particular, the participant who replaced 231% of familiarization sodium losses in SODIUM did so due to a sweat rate that was 42% greater and $[Na^+]_{sweat}$ 19% greater during familiarization compared with SODUIM. It is unclear why such a large discrepancy in sweat rate occurred, but given that running speed, T_{amb} and RH were the same during familiarization as SODIUM, the most likely explanation is an error in BM measurement or data recording. We note that removal of this outlier results in a mean discrepancy between familiarization and SODIUM for sweat rate of 12% and $[Na^+]_{sweat}$ of 17%, both close to the expected between-day variability based on previous literature.[22] Nonetheless, the individual variability in total sodium loss (3 participants >20% and 1 >40%) remains a potential concern for practitioners aiming to test sodium losses to plan sodium replacement strategies for a future day.

Practical Applications

Personalized sodium replacement during ~5-hour running does not stimulate increased ad libitum fluid intake and is unnecessary to optimize hydration status. Complete postexercise sodium replacement is also unnecessary for multiday events, due to urinary sodium conservation. Our findings under controlled laboratory conditions suggest that personalized sweat sodium testing and replacement are unlikely to be as precise as many athletes and practitioners expect.

Conclusions

Complete, personalized replacement of sweat sodium losses during 5-hour continuous running in the heat appears to have minimal impact on thirst, ad libitum water intake, net water balance, and body water shifts between compartments and no impact on thermophysiological strain. Furthermore, when water is consumed ad libitum, sodium replacement increases $[Na^+]_{plasma}$ and P_{Osm} rather than preventing its decline. There was no evidence for release of osmotically inactive sodium stores in response to a large sodium deficit incurred during running, when such a deficit is not accompanied by large reductions in total body water or plasma volume. Net sodium loss during exercise causes a substantial postexercise renal sodium conservation, although in ultraendurance exercise this is inadequate to fully restore sodium balance within 24 hours without deliberately increased postexercise sodium intake.

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